Who Connects the Professional World?

A Network Science Analysis of LinkedIn Job Titles

Analyzing the Structure of the Professional World via LinkedIn Connections

Do job titles co-occurring in the same company form a structured network and what does that structure reveal?

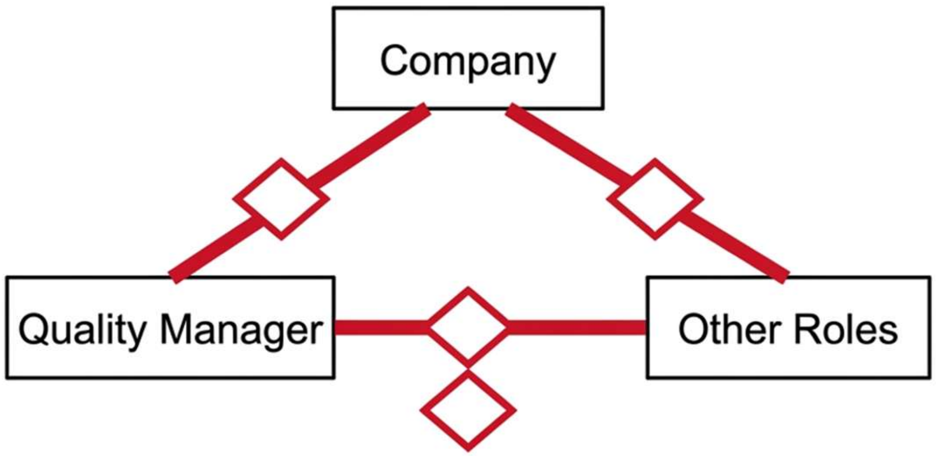
Which roles bridge different professional worlds?

Do roles cluster into functional communities?

How did the network evolve from 2022 to 2026?

1,863 raw connections → 1,736 cleaned → 1,301 companies → 393 job titles → 3,302 edges

Two Job Titles Get Connected Every Time They Share a Company



They get an edge in the Position to Position Network.

The Network Is Small-World: Clustering 22× Higher Than Random

Density:
0.0429
4.3% of possible edges exist

Giant Component:
225 nodes
57% of all roles are connected in one main structure

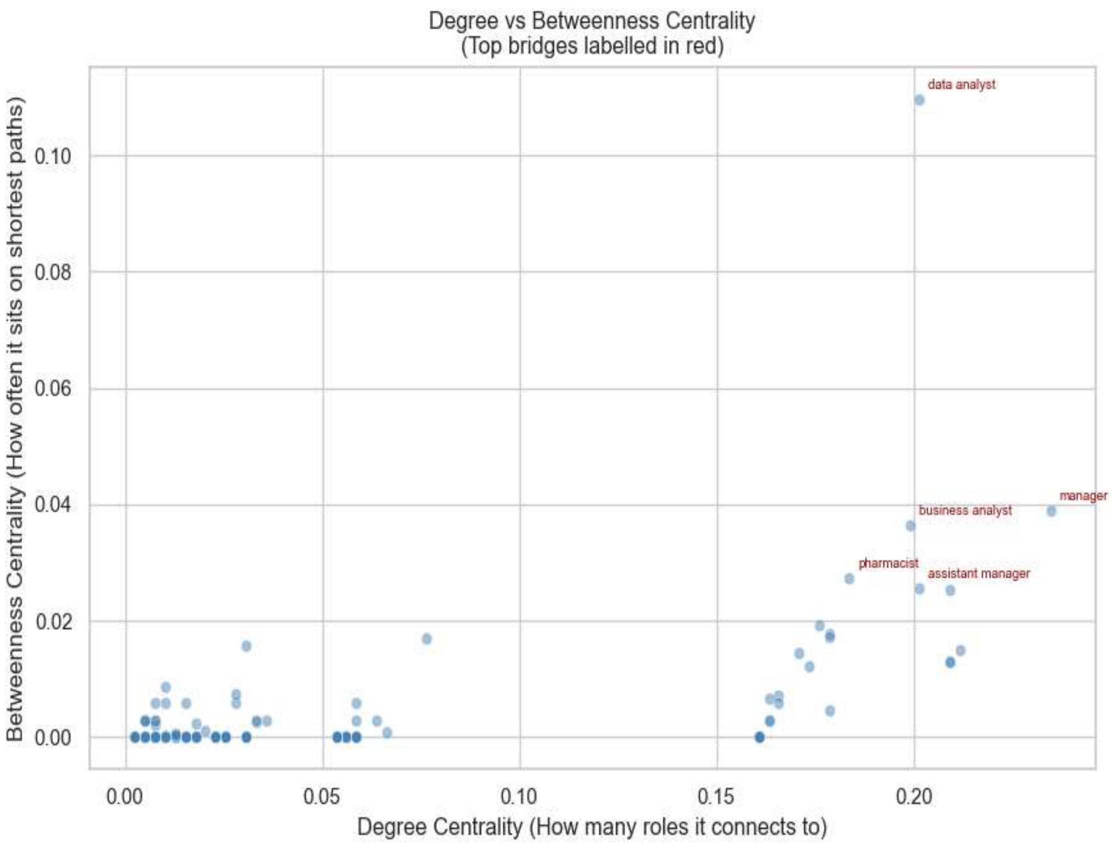
Avg Path Length:
2.61 hops
Six degrees of separation — achievable in just 3 hops

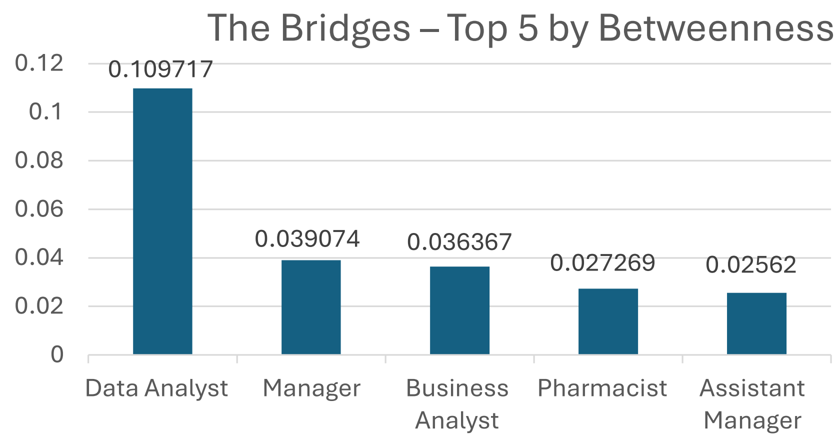
Clustering: 0.9028
22× higher than random baseline
Small World Confirmed ✓

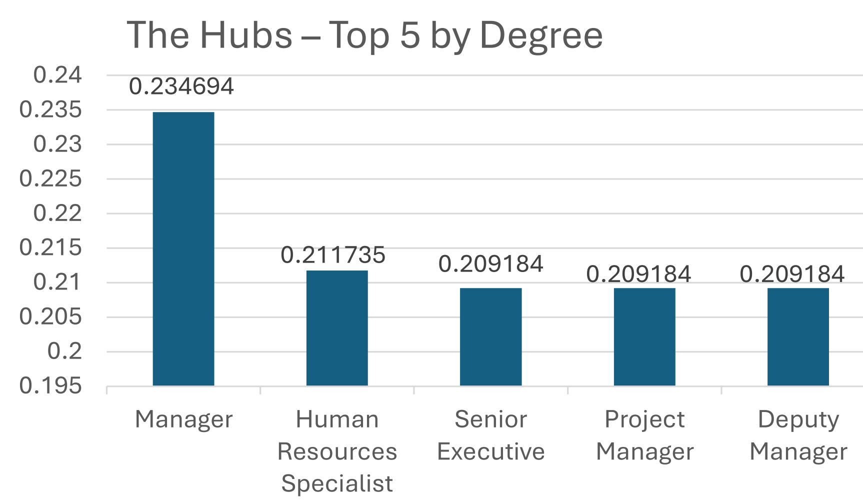
ER Clustering:
0.0416
What a purely random network would look like

Assortativity:
+0.6873
Positive value: Hub roles connect to other hubs (core-periphery)

Data Analyst Is the Most Powerful Bridge in the Entire Network







68 Communities Found All Interpretable Without Labels

Community 1 | 69 roles | Corporate Management & Analytics

Top roles: senior executive, assistant manager, business analyst, pharmacist, senior business analyst, senior data analyst

Community 2 | 55 roles | General Management & HR

Top roles: manager, human resources specialist, project manager, deputy manager, data analyst, it analyst

Community 3 | 38 roles | Data Science & Talent Acquisition

Top roles: analyst, data scientist, accounts receivable analyst, talent acquisition executive, deputy manager ai & automations

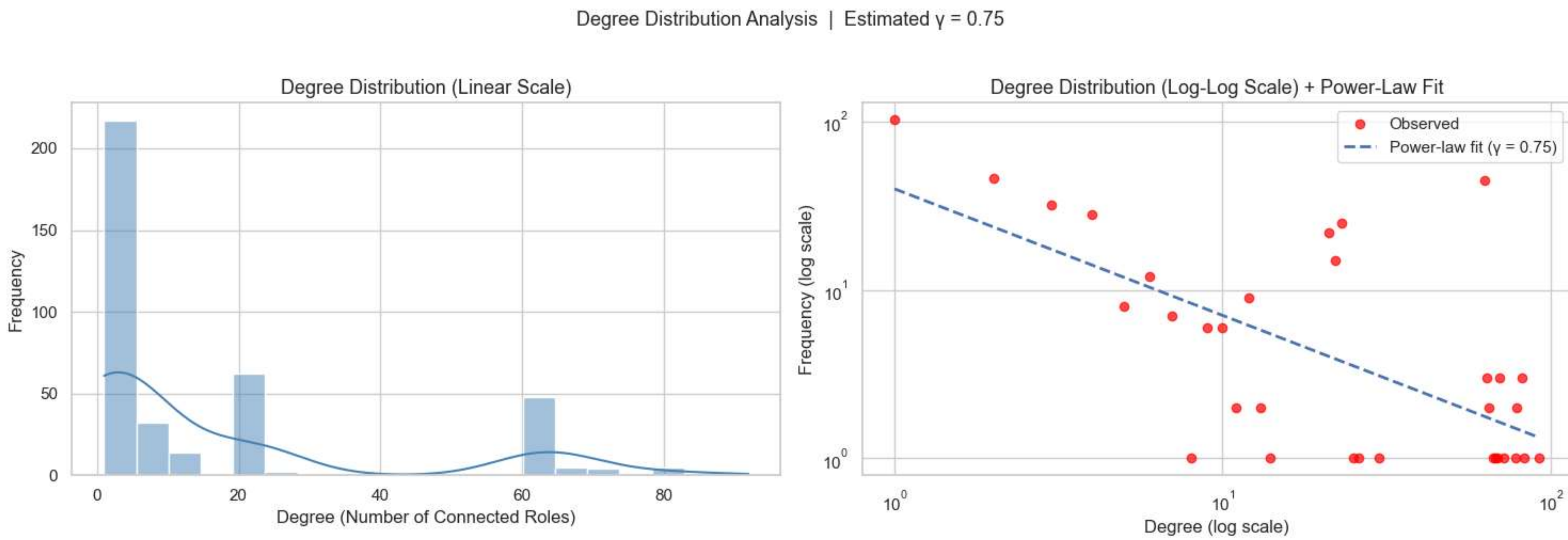
Community 4 | 27 roles | Academia & University

Top roles: assistant professor, graduate admissions advisor, its student technician, data analytics fellow, graduate student

Community 5 | 22 roles | Senior Leadership & HR Strategy

Top roles: vice president of strategic initiatives and workforce development, assistant director of human resources, director of human resources, acting president

The Network Is Scale Free With Extreme Hub Dominance (γ = 0.75)

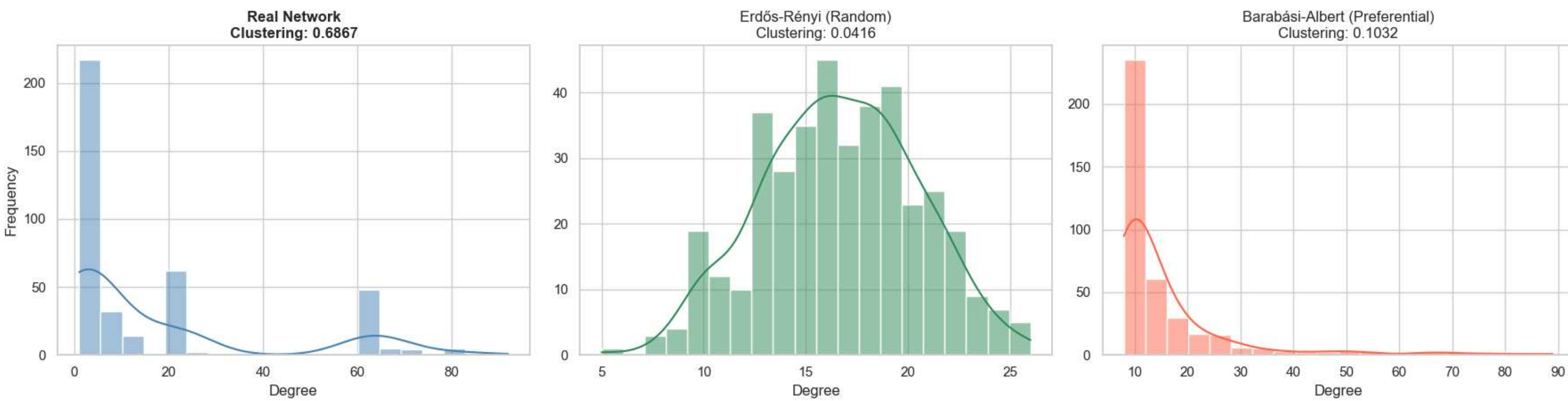


- Most job titles are specialists with few connections (peak at degree 1 5).
- A small number of generalist hubs completely dominate the topology (secondary cluster at degree 60+).
- Estimated γ = 0.75 — vastly below the typical 2–3 range, indicating highly extreme hub concentration.

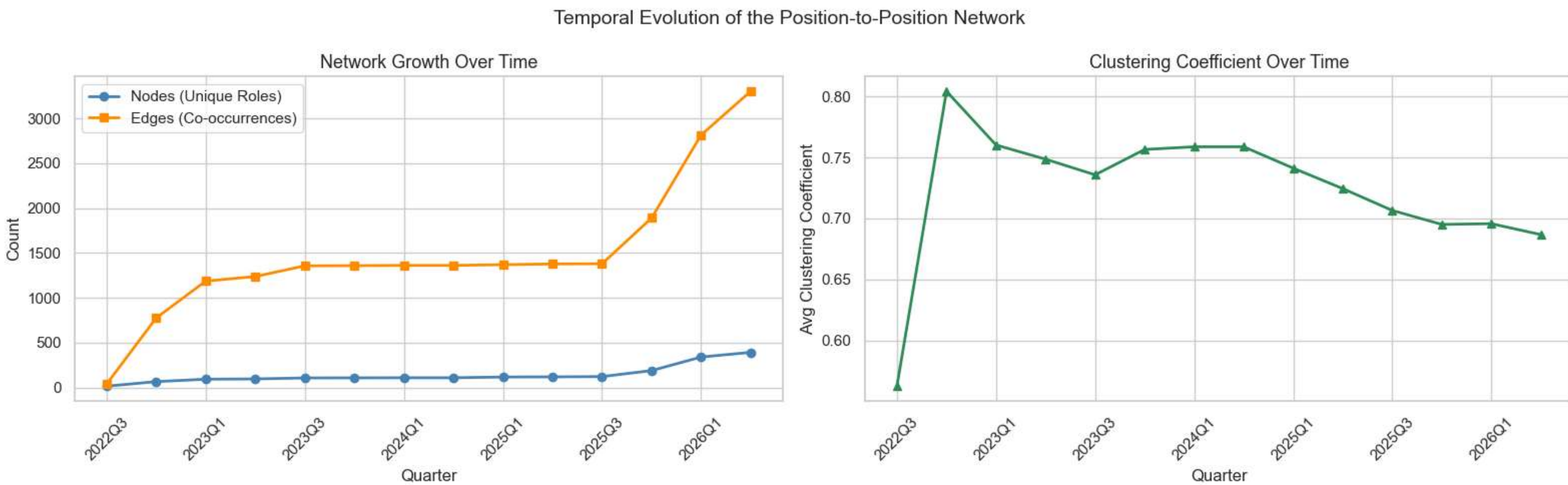
Real network's clustering is 16X ER and 7X BA

Metric	Real	ER	BA
Avg Path Length	2.61	2.43	2.40
Avg Clustering	0.6867	0.0416	0.1032
Density	0.0429	0.0424	0.0400

Degree Distribution: Real Network vs Theoretical Models



Edges Grew 8X Faster Than Nodes And Clustering Tells a Career Story



From 2022 to 2026, average Clustering fell from 0.81 to 0.69. The professional world is broadening across new sectors, rather than deepening into isolated silos.

The Professional World Has Structure — Network Science Can Read It

Checklist of Discoveries

- ✔ Small World confirmed (clustering 22× random baseline)
- ✔ Scale free distribution ($\gamma = 0.75$, extreme hub dominance)
- ✔ 68 functional communities — all organically interpretable
- ✔ Temporal densification signals broad career diversification

What Comes Next?

Scale: Test topology against datasets of thousands of users.

Normalize: Apply NLP to reduce 1,122 title variants to canonical roles.

Embed: Use Node2vec to build a predictive career path recommendation engine.

Predict: Utilize link prediction algorithms to identify hiring gaps in enterprise organizations.

**Your LinkedIn network is not just a contact list.
It is a mathematical map of the professional world.**